

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
library(tidyverse)
library(lubridate)
library(here)
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
Neonics <- read.csv(here("Data", "Raw", "ECOTOX_Neonicotinoids_Insects_raw.csv"),
  stringsAsFactors = TRUE)
Litter <- read.csv(here("Data", "Raw", "NEON_NIWO_Litter_massdata_2018-08_raw.csv"),
  stringsAsFactors = TRUE)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency’s ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: We are interested in the ecotoxicology of neonicotinoids on insects because they are widely used agricultural insecticides that can affect not only target pests but also non-target insects such as pollinators. Studying their effects helps us understand the potential ecological cascade — for example, declines in pollinators can disrupt plant reproduction, crop yields, and food webs.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: We are interested in studying litter and woody debris because they play a key role in nutrient cycling and soil fertility. As they decompose, they release essential nutrients back into the ecosystem and contribute to soil organic matter. In addition, decomposition rates act as indicators of environmental change, since they are strongly influenced by climate, soil conditions, and microbial activity. Monitoring these processes helps us understand how forests respond to changing environmental conditions.

4. How is litter and woody debris sampled as part of the NEON network? Read the `NEON_Litterfall_UserGuide.pdf` document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. NEON collects litter and woody debris in tower plots using 1–4 pairs of elevated and ground traps per plot. Trap locations are chosen randomly when woody cover exceeds 50%, or placed under vegetation when cover is sparse, with additional spacing rules to avoid features like roads, buildings, and streams. 2. The smallest unit of NEON litterfall sampling is an individual trap, each labeled with a unique `trapID`. Traps are nested within a hierarchy of identifiers: `trapID` → `subplotID` → `plotID` → `siteID` → `domainID`. Spatial information such as latitude, longitude, and elevation of plot centroids is provided along with uncertainty estimates. For higher accuracy, users can either apply the `geoNEON` R package or use NEON’s API together with a clip cell lookup table to determine exact trap locations. 3. NEON maintains data quality by applying automated validation checks during data entry and processing prior to release, along with ongoing revisions while data are provisional. Any corrections or updates are recorded in change logs, and known issues are flagged with a quality system—currently limited to a “`legacyData`” tag for older paper-based records.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics)
```

```
## [1] 4623 30
```

```
view(data_frame(Neonics))
```

```
## Warning: 'data_frame()' was deprecated in tibble 1.1.0.
## i Please use 'tibble()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

- Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
effect_freq <- sort(table(Neonics$Effect), decreasing = TRUE)
```

Answer: The most common effects studied are Population, Mortality, and Behavior. These are of particular interest because mortality measures direct toxic effects, behavior reflects sub-lethal changes that can still threaten survival, and population effects integrate these responses over time to show how neonicotinoid exposure influences long-term persistence of insect populations and the ecosystems they support.

- Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
species_freq <- summary(Neonics$Species.Common.Name, maxsum = 7)
species_freq
```

```
##           Honey Bee           Parasitic Wasp Buff Tailed Bumblebee
##           667           285           183
## Carniolan Honey Bee           Bumble Bee           Italian Honeybee
##           152           140           113
##           (Other)
##           3083
```

```
unique(Neonics$Species.Common.Name)
```

```
## [1] Coffee Bean Weevil           House Fly
## [3] Fall Armyworm                   Parastic Wasp
## [5] Parasitic Wasp                   Spiralling Whitefly
## [7] Mosquito                       Parasitic Wasp Family
## [9] Glasshouse Potato Wasp          Honey Bee
## [11] Carniolan Honey Bee             European Dark Bee
## [13] Bee                             Sweetpotato Whitefly
## [15] Buff Tailed Bumblebee           Pollen Beetle
## [17] Pear Sucker                     Cabbage Seedpod Weevil
## [19] Oblique Banded Leaf Roller     Plum Curculio
## [21] Southern House Mosquito         Beetle
## [23] Codling Moth                   San Jose Scale
## [25] Apple Aphid                    Minute Parasitic Wasps
```

## [27] Mirid Bug	European Apple Sawfly
## [29] Wasp Family	Insect Class
## [31] Speckled Cutworm Moth	Moth And Butterfly Order
## [33] Tarnished Plant Bug	European Tarnished Plant Bug
## [35] Braconid Wasp	Stingless Bee
## [37] European Red Mite	Lacewing
## [39] Pteromalid Wasp Family	Apple Maggot
## [41] Lady Beetle	Predatory Mite
## [43] Striped Cucumber Beetle	Two Spotted Lady Beetle
## [45] Alfalfa Plant Bug	Pistachio Psyllid
## [47] Wireworm	Ambrosia Beetle
## [49] Jassid	Beetle Family
## [51] Minute Pirate Bug Family	Strawberry Blossom Weevil
## [53] Gall Midge	Cotton Aphid
## [55] Red Scale Parasite	Asiatic Honey Bee
## [57] Spider/Mite Class	Araneoid Spider Order
## [59] Eulophid Wasp	Coconut Leaf Beetle
## [61] Long Horned Beetle Family	Aphid
## [63] Common Green Lacewing	Lacewing Family
## [65] Eleven-spotted Ladybird Beetle	Ladybird Beetle Family
## [67] Beetle Order	Banded Soft-winged Flower Beetle
## [69] Mealybug Destroyer	Snout Beetle Family, Weevil
## [71] Predaceous Fly	Meshweaver Spider
## [73] Oystershell Scale Parasitoid	Chalcid Wasp
## [75] Mite	Earwig
## [77] Western Flower Thrips	Common Thrip
## [79] Thrip	Western Bigeyed Bug
## [81] Chinch Bug	Asian Lady Beetle
## [83] Corn Earworm	Convergent Lady Beetle
## [85] Bee Order	Parasitoid
## [87] Western Plant Bug	Celer Crab Spider
## [89] Western Damsel Bug	Minute Pirate Bug
## [91] Minute Flour Bug	Hornfaced Bee
## [93] Turf Running-spider	Chilean Predatory Mite
## [95] Cabbage White	Diamondback Moth
## [97] Cotton Fleahopper	Jumping Spider Family
## [99] Red Imported Fire Ant	Whitemarked Fleahopper
## [101] Ladybird Beetle	Two-Spotted Spider Mite
## [103] Onion Thrip	Parasitoid Wasp
## [105] Cabbage Looper	Euonymus Scale
## [107] Ambrosia Bark Beetle	Fruit-tree Pinhole Borer
## [109] Black Twig Borer	Redbay Ambrosia Beetle
## [111] Asian Ambrosia Beetle	Assassin Bug
## [113] Fig Longicorn Beetle	Antlike Flower Beetle
## [115] Pine False Webworm	Blue Alfalfa Aphid
## [117] Pea Aphid	Ten-spot Ladybird Beetle
## [119] Hemlock Woolly Adelgid	Yellow Fever Mosquito
## [121] Encyrtid Parasitoid	Lined Click Beetle
## [123] Ichneumonid Wasp	Egg Parasitoid
## [125] Fairyfly	Fairyfly Parasitoid
## [127] Fruit Fly	Pepper Weevil
## [129] Southern One-Year Canegrub	Childers Canegrub
## [131] California Red Scale	Aphid Family
## [133] Soybean Aphid	Spirea Aphid

## [135] Dwarf Honey Bee	European Honey Bee
## [137] Italian Honeybee	Macedonian Honey Bee
## [139] Rose Sawfly	Scarab Beetle
## [141] Pygmy Mangold Beetle	Leaf Cutting Ant
## [143] Encyrtid Wasp	Wasp
## [145] Silverleaf Whitefly	Bumble Bee
## [147] Buff-tailed Bumblebee	Bumblebee
## [149] Yellow-faced Bumblebee	Silkworm
## [151] Cabbage Aphid	Horned Oak Gall Wasp
## [153] Ground Beetle Family	Ectoparasitoid Wasp
## [155] Mediterranean Fruit Fly	Cactus Lady Beetle
## [157] Centipede Class	Flatheaded Appletree Borer
## [159] Leaf Beetle Family	Sevenspotted Lady Beetle
## [161] Citricola Scale	Formosan Subterranean Termite
## [163] Hawthorn Lace Bug	Green June Beetle
## [165] Spotless Ladybird Beetle	Spotted Sunflower Stem Weevil
## [167] Cabbage Root Fly	Longtailed Fruit Fly Parasite
## [169] Asian Citrus Psyllid	Fly/Mosquito/Midge Order
## [171] Soldier Beetle	Russian Wheat Aphid
## [173] Painted Maple Aphid	Potato Leafhopper
## [175] Hover Fly	Tobacco Flea Beetle
## [177] Sugarcane Beetle	Calico Scale
## [179] Predacious Mite	Birch Leafminer
## [181] Ant Family	Tobacco Thrip
## [183] Mulberry Pyralid	Spider
## [185] Budworm	True Bug Order
## [187] Black-spotted Lady Beetle	Hister Beetle Family
## [189] Glassy-winged Sharpshooter	Scale Insect Order
## [191] Linyphiid Spider	Egyptian Alfalfa Weevil
## [193] Cottony Cushion Sale	Delphacid Planthopper
## [195] Tooth-necked Fungus Beetle	Ant
## [197] Negatoria Canegrub	Colorado Potato Beetle
## [199] Argentine Ant	Sheet-Web Weaver Family
## [201] Turnip Aphid	Serpentine Leafminer
## [203] Wolf Spider Family	Plant Bug
## [205] Aphid Wasp	Alfalfa Leafcutter Bee
## [207] Neotropical Stingless Bee	Mite Order
## [209] Rose Grain Aphid	Braconid Parasitoid Wasp
## [211] Tobacco Aphid	Green Peach Aphid
## [213] Green Rice Leafhopper	Clouded Plant Bug
## [215] Brown Planthopper	Sap Beetle Family
## [217] Moth Family	Alkali Bee
## [219] Maple Spider Mite	Eucalyptus Gall Wasp
## [221] Harvestman Spider Order	Oribatid Mite Suborder
## [223] Beetle Mite Family	Predatory Bug
## [225] Predator Bug	Grasshopper/Cricket/Locust Order
## [227] Mason Bee	Black Vine Weevil
## [229] Rove Beetle	Citrus Red Mite
## [231] Pond Wolf Spider	Brown Scale
## [233] Banana Aphid	Potato Tuberworm
## [235] Citrus Leafminer	Spotted Tentiform Leafminer
## [237] Crucifer Flea Beetle	Flea Beetle
## [239] Mite Family	Spined Soldier Bug
## [241] Japanese Beetle	Hawthorn Leaf Miner

## [243]	Obscure Mealybug	Braconid Parasitoid
## [245]	Erythrina Gall Wasp	Subterranean Termite
## [247]	Eastern Subterranean Termite	Vedalia Beetle
## [249]	Tick/Chigger/Mite Order	Hemlock Woolly Adelgid Lady Beetle
## [251]	Tawny Mole Cricket	Scarab Beetle Family
## [253]	Cardamom Thrip	Garden Symphytan
## [255]	English Grain Aphid	Pea And Bean Weevil
## [257]	White-backed Planthopper	Moth
## [259]	Southern Army Worm	Beet Armyworm
## [261]	Egyptian Cotton Leafworm	Rove Beetle Family
## [263]	Azalea Lace Bug	Spider Mite Destroyer
## [265]	Ox Beetle	Pseudocentipede Class
## [267]	Parsnip Seed Wasp	Eulophid Parasitoid
## [269]	Yellow Mealworm Beetle	Large-Jawed Orb Weaver Family
## [271]	Thrip Family	Thrip Order
## [273]	Crapemyrtle Aphid	Spring Tiphia
## [275]	Sugarcane Grub	Black Citrus Aphid
## [277]	Greenhouse Whitefly	White Apple Leafhopper Nymph
## [279]	Tenebrionid Beetle	Stink Bug
## [281]	Cutworm	Oriental Beetle
## [283]	Rice Leaf Folder Moth	Banded Sunflower Moth
## [285]	Sunflower Midge	Brown Stinkbug
## [287]	Scelionid Wasp	Sunflower Moth
## [289]	Carrot Weevil	Ground Beetle
## [291]	Red Sunflower Seed Weevil	Grey Sunflower Seed Weevil
## [293]	Weevil	Twicestabbed Lady Beetle
## [295]	Scale Picnic Beetle	Armoured Scale Family
## [297]	Engraver Beetle	Shiny Spider Beetle
## [299]	Minute Lady Beetles	Damselbug Family
## [301]	Bee Family	Monarch Butterfly
## [303]	Onion Maggot	
## 303 Levels:	Alfalfa Leafcutter Bee Alfalfa Plant Bug ... Yellow-faced Bumblebee	

```
table(Neonics$Species.Common.Name)
```

##		
##	Alfalfa Leafcutter Bee	Alfalfa Plant Bug
##	8	2
##	Alkali Bee	Ambrosia Bark Beetle
##	2	3
##	Ambrosia Beetle	Ant
##	6	4
##	Ant Family	Antlike Flower Beetle
##	9	1
##	Aphid	Aphid Family
##	2	38
##	Aphid Wasp	Apple Aphid
##	6	5
##	Apple Maggot	Araneoid Spider Order
##	9	17
##	Argentine Ant	Armoured Scale Family
##	21	13
##	Asian Ambrosia Beetle	Asian Citrus Psyllid
##	3	60

##	Asian Lady Beetle	Asiatic Honey Bee
##	76	9
##	Assassin Bug	Azalea Lace Bug
##	2	2
##	Banana Aphid	Banded Soft-winged Flower Beetle
##	2	1
##	Banded Sunflower Moth	Bee
##	1	8
##	Bee Family	Bee Order
##	1	17
##	Beet Armyworm	Beetle
##	1	21
##	Beetle Family	Beetle Mite Family
##	3	7
##	Beetle Order	Birch Leafminer
##	47	3
##	Black Citrus Aphid	Black Twig Borer
##	1	3
##	Black Vine Weevil	Black-spotted Lady Beetle
##	6	18
##	Blue Alfalfa Aphid	Braconid Parasitoid
##	1	12
##	Braconid Parasitoid Wasp	Braconid Wasp
##	3	33
##	Brown Planthopper	Brown Scale
##	5	2
##	Brown Stinkbug	Budworm
##	2	2
##	Buff Tailed Bumblebee	Buff-tailed Bumblebee
##	183	39
##	Bumble Bee	Bumblebee
##	140	8
##	Cabbage Aphid	Cabbage Looper
##	2	38
##	Cabbage Root Fly	Cabbage Seedpod Weevil
##	1	4
##	Cabbage White	Cactus Lady Beetle
##	2	1
##	Calico Scale	California Red Scale
##	18	3
##	Cardamom Thrip	Carniolan Honey Bee
##	2	152
##	Carrot Weevil	Celer Crab Spider
##	2	2
##	Centipede Class	Chalcid Wasp
##	2	25
##	Childers Canegrub	Chilean Predatory Mite
##	6	8
##	Chinch Bug	Citricola Scale
##	7	2
##	Citrus Leafminer	Citrus Red Mite
##	23	1
##	Clouded Plant Bug	Coconut Leaf Beetle
##	2	6

##	Codling Moth	Coffee Bean Weevil
##	19	2
##	Colorado Potato Beetle	Common Green Lacewing
##	57	4
##	Common Thrip	Convergent Lady Beetle
##	12	25
##	Corn Earworm	Cotton Aphid
##	14	33
##	Cotton Fleahopper	Cottony Cushion Sale
##	2	1
##	Crapemyrtle Aphid	Crucifer Flea Beetle
##	1	3
##	Cutworm	Damselbug Family
##	3	1
##	Delphacid Planthopper	Diamondback Moth
##	3	13
##	Dwarf Honey Bee	Earwig
##	8	5
##	Eastern Subterranean Termite	Ectoparasitoid Wasp
##	12	1
##	Egg Parasitoid	Egyptian Alfalfa Weevil
##	17	2
##	Egyptian Cotton Leafworm	Eleven-spotted Ladybird Beetle
##	3	6
##	Encyrtid Parasitoid	Encyrtid Wasp
##	3	6
##	English Grain Aphid	Engraver Beetle
##	1	2
##	Erythrina Gall Wasp	Eucalyptus Gall Wasp
##	49	4
##	Eulophid Parasitoid	Eulophid Wasp
##	9	13
##	Euonymus Scale	European Apple Sawfly
##	75	4
##	European Dark Bee	European Honey Bee
##	66	4
##	European Red Mite	European Tarnished Plant Bug
##	6	4
##	Fairyfly	Fairyfly Parasitoid
##	1	18
##	Fall Armyworm	Fig Longicorn Beetle
##	6	2
##	Flatheaded Appletree Borer	Flea Beetle
##	20	1
##	Fly/Mosquito/Midge Order	Formosan Subterranean Termite
##	3	3
##	Fruit Fly	Fruit-tree Pinhole Borer
##	6	3
##	Gall Midge	Garden Symphylan
##	1	4
##	Glasshouse Potato Wasp	Glassy-winged Sharpshooter
##	10	2
##	Grasshopper/Cricket/Locust Order	Green June Beetle
##	1	5

##	Green Peach Aphid	Green Rice Leafhopper
##	14	3
##	Greenhouse Whitefly	Grey Sunflower Seed Weevil
##	1	1
##	Ground Beetle	Ground Beetle Family
##	3	27
##	Harvestman Spider Order	Hawthorn Lace Bug
##	1	2
##	Hawthorn Leaf Miner	Hemlock Woolly Adelgid Lady Beetle
##	1	16
##	Hemlock Woolly Adelgid	Hister Beetle Family
##	16	2
##	Honey Bee	Horned Oak Gall Wasp
##	667	20
##	Hornfaced Bee	House Fly
##	5	14
##	Hover Fly	Ichneumonid Wasp
##	6	3
##	Insect Class	Italian Honeybee
##	17	113
##	Japanese Beetle	Jassid
##	94	12
##	Jumping Spider Family	Lacewing
##	2	10
##	Lacewing Family	Lady Beetle
##	9	18
##	Ladybird Beetle	Ladybird Beetle Family
##	23	30
##	Large-Jawed Orb Weaver Family	Leaf Beetle Family
##	3	20
##	Leaf Cutting Ant	Lined Click Beetle
##	3	2
##	Linyphiid Spider	Long Horned Beetle Family
##	4	5
##	Longtailed Fruit Fly Parasite	Macedonian Honey Bee
##	1	7
##	Maple Spider Mite	Mason Bee
##	2	22
##	Mealybug Destroyer	Mediterranean Fruit Fly
##	9	3
##	Meshweaver Spider	Minute Flour Bug
##	2	3
##	Minute Lady Beetles	Minute Parasitic Wasps
##	1	18
##	Minute Pirate Bug	Minute Pirate Bug Family
##	62	2
##	Mirid Bug	Mite
##	18	16
##	Mite Family	Mite Order
##	3	12
##	Monarch Butterfly	Mosquito
##	13	22
##	Moth	Moth And Butterfly Order
##	7	17

##	Moth Family	Mulberry Pyralid
##	3	18
##	Negatoria Canegrub	Neotropical Stingless Bee
##	3	8
##	Oblique Banded Leaf Roller	Obscure Mealybug
##	6	6
##	Onion Maggot	Onion Thrip
##	4	16
##	Oribatid Mite Suborder	Oriental Beetle
##	6	4
##	Ox Beetle	Oystershell Scale Parasitoid
##	14	17
##	Painted Maple Aphid	Parasitic Wasp
##	1	285
##	Parasitic Wasp Family	Parasitoid
##	8	30
##	Parasitoid Wasp	Parastic Wasp
##	51	58
##	Parsnip Seed Wasp	Pea And Bean Weevil
##	4	4
##	Pea Aphid	Pear Sucker
##	12	4
##	Pepper Weevil	Pine False Webworm
##	1	1
##	Pistachio Psyllid	Plant Bug
##	6	1
##	Plum Curculio	Pollen Beetle
##	5	1
##	Pond Wolf Spider	Potato Leafhopper
##	12	20
##	Potato Tuberworm	Predaceous Fly
##	7	2
##	Predacious Mite	Predator Bug
##	1	1
##	Predatory Bug	Predatory Mite
##	13	33
##	Pseudocentipede Class	Pteromalid Wasp Family
##	1	1
##	Pygmy Mangold Beetle	Red Imported Fire Ant
##	2	4
##	Red Scale Parasite	Red Sunflower Seed Weevil
##	14	1
##	Redbay Ambrosia Beetle	Rice Leaf Folder Moth
##	6	1
##	Rose Grain Aphid	Rose Sawfly
##	1	2
##	Rove Beetle	Rove Beetle Family
##	5	27
##	Russian Wheat Aphid	San Jose Scale
##	7	5
##	Sap Beetle Family	Scale Insect Order
##	3	3
##	Scale Picnic Beetle	Scarab Beetle
##	1	29

##	Scarab Beetle Family	Scelionid Wasp
##	3	5
##	Serpentine Leafminer	Sevenspotted Lady Beetle
##	2	46
##	Sheet-Web Weaver Family	Shiny Spider Beetle
##	3	1
##	Silkworm	Silverleaf Whitefly
##	18	6
##	Snout Beetle Family, Weevil	Soldier Beetle
##	47	7
##	Southern Army Worm	Southern House Mosquito
##	1	10
##	Southern One-Year Canegrub	Soybean Aphid
##	7	6
##	Speckled Cutworm Moth	Spider
##	5	3
##	Spider Mite Destroyer	Spider/Mite Class
##	2	24
##	Spined Soldier Bug	Spiralling Whitefly
##	14	8
##	Spirea Aphid	Spotless Ladybird Beetle
##	1	11
##	Spotted Sunflower Stem Weevil	Spotted Tentiform Leafminer
##	1	2
##	Spring Tiphia	Stingless Bee
##	29	25
##	Stink Bug	Strawberry Blossom Weevil
##	2	1
##	Striped Cucumber Beetle	Subterranean Termite
##	4	6
##	Sugarcane Beetle	Sugarcane Grub
##	4	3
##	Sunflower Midge	Sunflower Moth
##	1	1
##	Sweetpotato Whitefly	Tarnished Plant Bug
##	37	7
##	Tawny Mole Cricket	Ten-spot Ladybird Beetle
##	2	1
##	Tenebrionid Beetle	Thrip
##	3	6
##	Thrip Family	Thrip Order
##	5	29
##	Tick/Chigger/Mite Order	Tobacco Aphid
##	2	27
##	Tobacco Flea Beetle	Tobacco Thrip
##	24	1
##	Tooth-necked Fungus Beetle	True Bug Order
##	20	45
##	Turf Running-spider	Turnip Aphid
##	2	2
##	Twicestabbed Lady Beetle	Two Spotted Lady Beetle
##	1	10
##	Two-Spotted Spider Mite	Vedalia Beetle
##	6	18

```
##           Wasp           Wasp Family
##           4           1
##           Weevil       Western Bigeyed Bug
##           1           2
##           Western Damsel Bug       Western Flower Thrips
##           2           15
##           Western Plant Bug       White Apple Leafhopper Nymph
##           2           2
##           White-backed Planthopper       Whitemarked Fleahopper
##           2           2
##           Wireworm       Wolf Spider Family
##           69           4
##           Yellow Fever Mosquito       Yellow Mealworm Beetle
##           13           1
##           Yellow-faced Bumblebee
##           4
```

```
top6_species <- head(sort(table(Neonics$Species.Common.Name), decreasing = TRUE), 6)
head(top6_species)
```

```
##
##           Honey Bee           Parasitic Wasp Buff Tailed Bumblebee
##           667           285           183
##           Carniolan Honey Bee           Bumble Bee           Italian Honeybee
##           152           140           113
```

Answer: The six most commonly studied species in the dataset were Honey Bee, Parasitic Wasp Buff, Tailed Bumblebee, Carniolan Honey Bee, Bumble Bee and Italian Honeybee. They are studied because most are pollinators essential for crops and ecosystems, while parasitoid wasps act as natural pest control. Their health directly impacts food security, biodiversity, and agriculture, making them more important than other insects.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$'Conc.1..Author.')
```

```
## [1] "factor"
```

Answer: The class of the `Conc.1..Author.` column is factor. It is not numeric because some entries include text or non-numeric symbols (detection limits, units, or missing value codes), which prevent R from storing the data as purely numeric values.

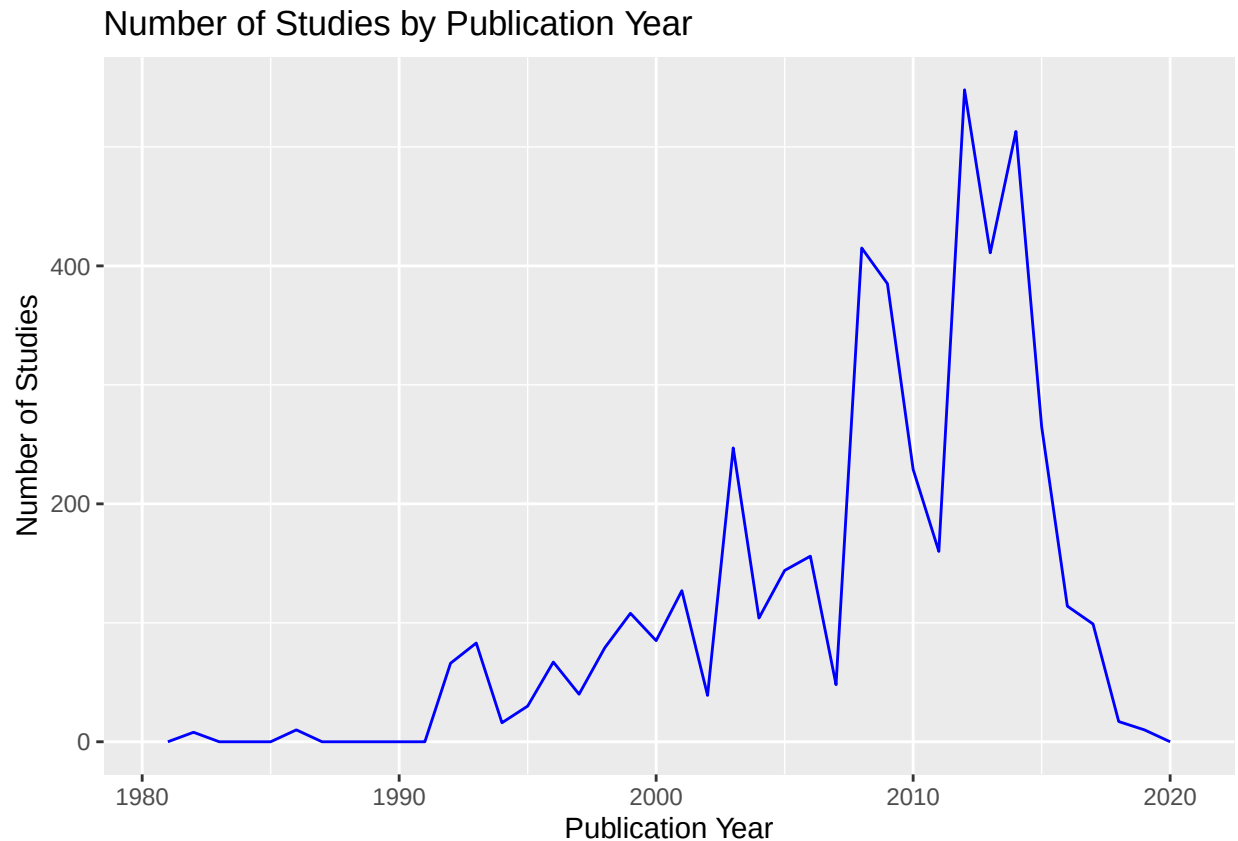
Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
library(ggplot2)

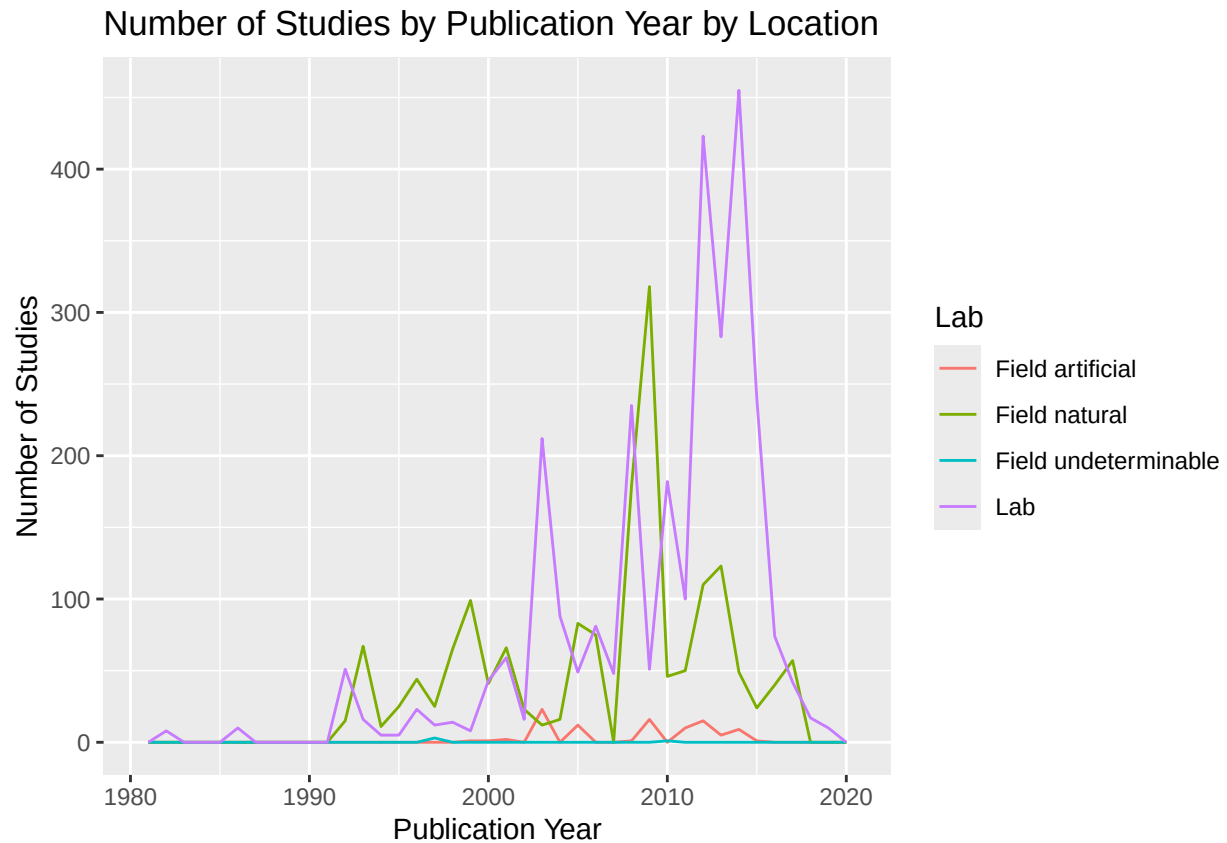
ggplot(data = Neonics, aes(x = Publication.Year)) +
```

```
geom_freqpoly(binwidth = 1, color = "blue") +
labs(
  title = "Number of Studies by Publication Year",
  x = "Publication Year",
  y = "Number of Studies"
)
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(data = Neonics, aes(x = Publication.Year, color= Test.Location)) +
geom_freqpoly(binwidth = 1) +
labs(
  title = "Number of Studies by Publication Year by Location",
  x = "Publication Year",
  y = "Number of Studies",
  color = Neonics$Test.Location
)
```



Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test locations are Lab and Field Natural. Lab studies dominate, especially after the early 2000s, while Field Natural studies are also frequent but show more variability across years, it's most common around 2008. Other locations, such as Field Artificial and Field Undeterminable, are relatively uncommon throughout the time series.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(Neonics, aes(x = Endpoint)) +
  geom_bar(fill = "purple") +
  labs(
    title = "Counts of Endpoints in ECOTOX Neonics Dataset",
    x = "Endpoint",
    y = "Count"
  ) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```

Counts of Endpoints in ECOTOX Neonics Dataset



Answer: The two most common end points are LOEL and NOEL. LOEL is in Terrestrial database, defined as Lowest -observable -effect-level: lowest dose (concentration) producing effects that were significantly different (as reported by authors) from responses of controls (LOEAL/LOEC). NOEL: is in Terrestrial database, Defined as No-observable-effect-level: highest dose (concentration) producing effects not significantly different from responses of controls according to author's reported statistical test (NOEAL/NOEC)

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
view(data_frame(Litter))
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
Litter$collectDate <- as.Date(as.character(Litter$collectDate), format = "%Y-%m-%d")
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

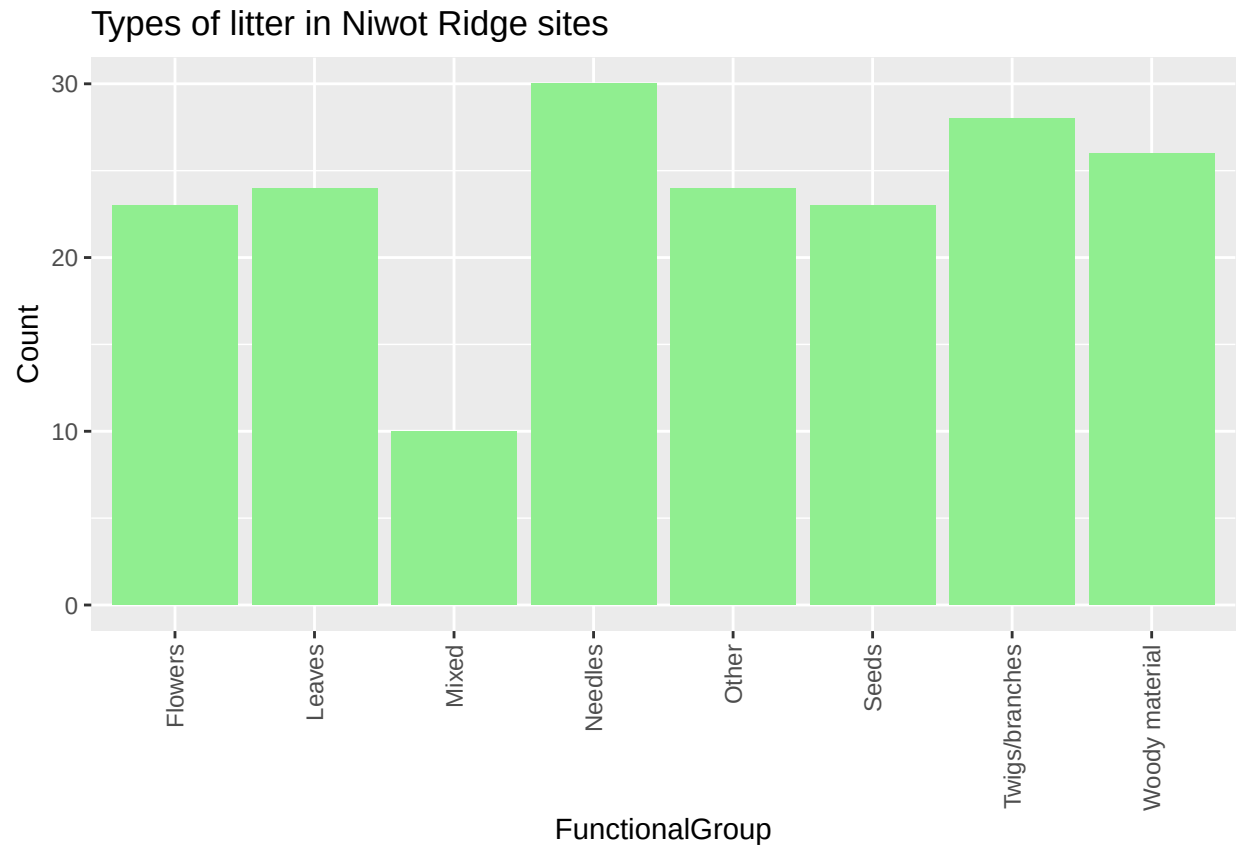
```
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061  
##      20      19      18      15      14       8      16      17  
## NIWO_062 NIWO_063 NIWO_064 NIWO_067  
##      14      14      16      17
```

Answer: `unique(Litter$plotID)` lists each distinct plotID sampled at Niwot Ridge once, without repetition, while `summary(Litter$plotID)` shows the frequency of observations for each plotID, providing counts at the same time.

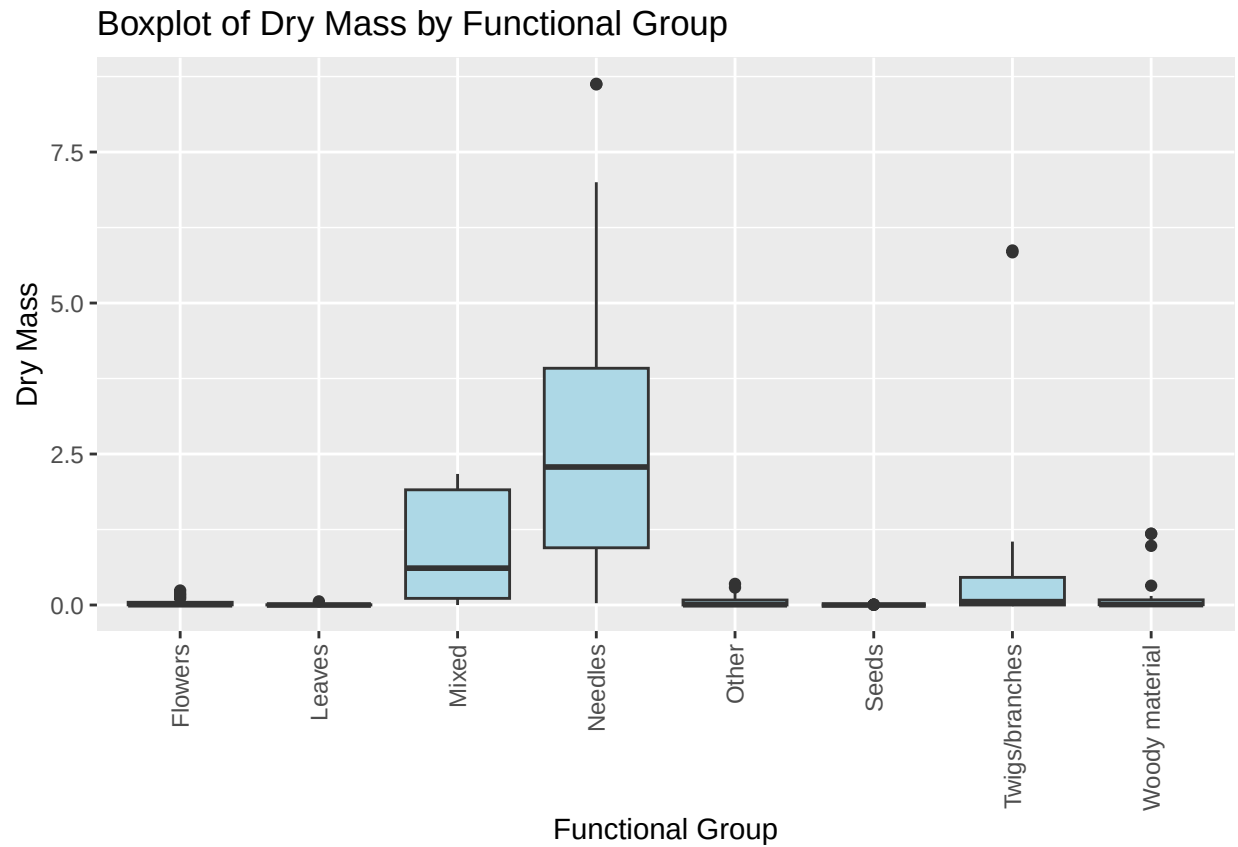
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(Litter, aes(x = functionalGroup)) +  
  geom_bar(fill = 'lightgreen') +  
  labs(  
    title = "Types of litter in Niwot Ridge sites",  
    x = "FunctionalGroup",  
    y = "Count"  
  ) +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```

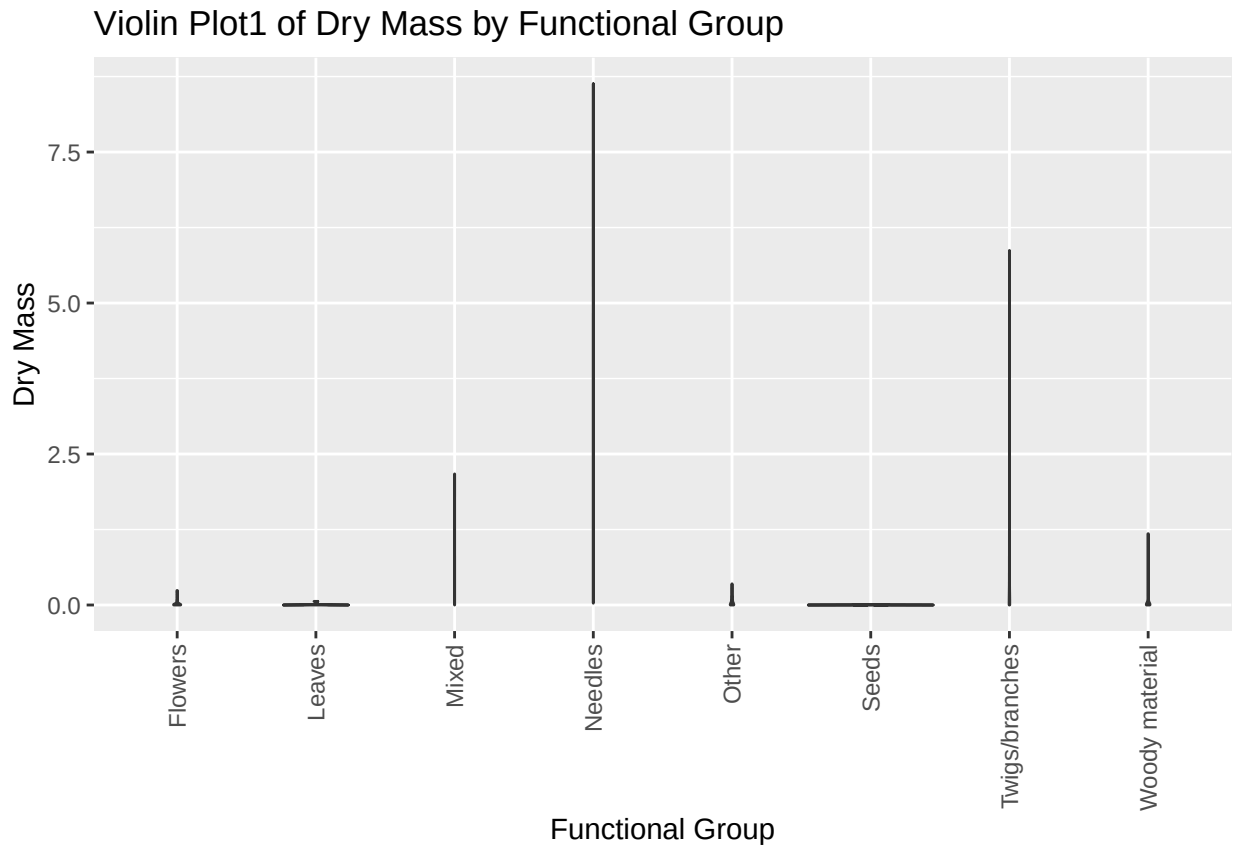



15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +
  geom_boxplot(fill = "lightblue") +
  labs(
    title = "Boxplot of Dry Mass by Functional Group",
    x = "Functional Group",
    y = "Dry Mass"
  ) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



```
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +
  geom_violin(fill = "lightblue") +
  labs(
    title = "Violin Plot1 of Dry Mass by Functional Group",
    x = "Functional Group",
    y = "Dry Mass"
  ) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: A boxplot is more effective than a violin plot in this case because dryMass values are highly skewed, with a few very large outliers. The boxplot clearly shows medians, quartiles, and outliers, while the violin plot becomes distorted and uninformative when extreme values stretch the distribution.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Litters with Needles tend to have the highest biomass at these sites.